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26 July 2026

Airbus

Hiring Team

Application: Master Thesis on AI Suitability Evaluation for Modelica Physical Models in Hamburg

Dear Hiring Team,

I am applying for the Master Thesis position on AI Suitability Evaluation for Modelica Physical Models at Airbus in Hamburg. The brief on evaluating the broad feasibility, accuracy, and engineering suitability of local open weights Large Language Models for physical modelling in Modelica, and on comparing the results against reference implementations, maps directly to the LLM system building and honest ML benchmarking work I have been shipping over the past year.

My Hybrid RAG Orchestrator is a working LLM system that classifies user intent into three execution paths, local knowledge retrieval, external web search, and direct conversational logic, built on Llama 3.1 8b via Groq, LangChain orchestration, a persistent ChromaDB vector store with HuggingFace MiniLM L6 v2 embeddings, and a stateful memory agent shipped behind a Streamlit interface. That gives me the concrete grounding in open weight LLM behaviour, prompt design, and end to end LLM engineering that this thesis needs.

At eRay GmbH I built an end to end recursive time series pipeline forecasting four water quality indicators and benchmarked six models head to head including Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, using CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals for decision support under uncertainty. I enforced strict anti leakage rules across the pipeline and surfaced the honest finding that some indicators are physically predictable while others are not. That mindset carries directly into evaluating whether a local LLM is genuinely suitable for engineering physics modelling or is merely producing plausible looking outputs.

In CreditIQ I built rigorous evaluation harnesses with SHAP driven subgroup analysis, cut the false negative rate from 44 percent to 16.7 percent while holding accuracy at 75 percent, and backed the pipeline with unit tests at 100 percent branch coverage. My bachelor thesis compares six classifiers on a clinical dataset with 10 fold cross validation, chooses ROC AUC over accuracy on an imbalanced dataset, and is written up as an IEEE style paper with a candid limitations section. I hold the NVIDIA Building LLM Applications With Prompt Engineering, AWS Academy Cloud Foundations, and Google Data Analytics Foundations certificates. I would be glad to discuss the exact Modelica reference set and evaluation criteria with the Hamburg team.

Kind regards,

Rahul Rawat